Separating decision-rule failures from representation failures in external validation of diabetic retinopathy grading models

Ahmadreza Nourozi[a,*], Alireza Chegeni[b]

[a] *Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Schlossplatz 4, 91054 Erlangen, Germany*

[b] *School of Electrical and Computer Engineering, College of Engineering, University of Tehran, Tehran, Iran*

Abstract

Failures of diabetic retinopathy (DR) grading models on new populations are usually blamed on the learned representation, although ordinal models turn scores into grades with a rarely examined decision rule. Under analysis plans committed in advance, we evaluated ten development models on four external populations (APTOS 2019, DDR, DeepDRiD, SUSTech-SYSU), a reverse-direction model, two models fine-tuned on 28,096 EyePACS images, five seeds and a published model's released predictions, separating per-threshold discrimination from the score-to-grade mapping. Referral ranking largely transferred (area under the curve at least 0.914), but the score given the true grade moved in opposite directions: the same models over-graded APTOS and under-graded DDR, so a threshold fitted for 85% development sensitivity reached 99.33% and 63.38%. Fitted on EyePACS instead, the threshold over-referred everywhere (90.8–100.0%): the direction of the shift belonged to the population, the sign of the threshold error to the development labels. Refitting cut-points recovered mild-grade recall on APTOS and reverse-

*Corresponding author.

Email address: Ahmadreza.nourozi@fau.de (Ahmadreza Nourozi)

direction referral sensitivity (43.5% to 77.9%) but not lost ranking; label-free prior adaptation did not help. Contrary to two predictions, models differed more than deployable rules, and the collapse of mild recall was specific to one corpus, although mild F1 stayed low everywhere. The risk that a locally set threshold misses an 80% sensitivity floor depended on the number of referable cases labelled, not of images: about 120 kept it near one in ten (about 1 000 images at 10% prevalence). External evaluations should report per-threshold discrimination and operating curves, and referral thresholds should be set locally.

Keywords: diabetic retinopathy, ordinal classification, decision threshold, external validation, dataset shift, recalibration, deep learning

1. Introduction

Diabetic retinopathy (DR) affects about one in five people with diabetes, an estimated 103 million adults in 2020 and a projected 161 million by 2045[1]. Deep learning systems that grade fundus photographs reached specialist-level agreement in development studies[2, 3], and autonomous screening devices have been authorised on the strength of prospective pivotal trials[4, 5]. The same class of system behaves very differently once it leaves the population it was built on: seven commercial algorithms applied to the same real-world images ranged from 51% to 86% in referable-disease sensitivity[6], and three algorithms evaluated against one reference standard in Indian public-health clinics ranged from 14% to 96% in specificity[7].

Such failures are usually read as statements about what the network has learned. A recent frozen-encoder benchmark found that two of three encoders,

including the retinal foundation model RETFound, did not identify a single mild case in an external corpus (grade-1 F1 of 0.000) and attributed the result to the encoders' representations[8]. An ordinal grading model does not output a grade, however. It outputs scores, and a *decision rule* — a set of cut-points — turns them into a grade and a referral. The common default, 0.5 on every cumulative output or an argmax, is a choice that is rarely reported or examined. If a model still ranks images correctly but its cut-points sit in the wrong place for a new population, it will look blind to a grade it in fact separates, and two models that differ mainly in where their defaults fall will look as if they differ in what they learned.

Parts of this idea are not new. Recalibration before deployment is standard advice for clinical prediction models[9, 10]. ORDER-DR, an ordinal DR model developed on APTOS, explicitly distinguished threshold-dependent grade assignment from threshold-independent risk ranking and noted that models trained on one public dataset can show threshold shift externally[11]. For binary referable DR across fundus cameras, a recent preprint showed that discrimination transfers while probability calibration degrades, mostly through prevalence, and that 100–200 labelled target images restore it [12]; a public model repository reports that calibration degrades unevenly across ordinal thresholds under domain shift[13]. What these studies do not provide is a decomposition across many models and several populations of how much of an external failure, and of an apparent difference between models, the ordinal decision rule accounts for; evidence on why the score-to-grade mapping moves; a rule for comparing models that removes it; and the label cost and risk of repairing it.

We provide these in a study whose analysis plans were committed to version control before the corresponding analyses were run, reporting falsified predictions with the same weight as confirmed ones. Ten development models trained on 2,260 images were evaluated on four external populations — APTOS 2019 (India), DDR (China, 147 hospitals), DeepDRiD (Shanghai) and SUSTech-SYSU (Shenzhen) — and a model trained on APTOS was evaluated in the reverse direction. Two further models were fine-tuned on 28,096 EyePACS images instead of the 2,260-image development pool, to test whether the findings depend on the development set and its labels; five seeds of the reference model measured training noise, and the released predictions of ORDER-DR were reanalysed. The contributions are:

- a decomposition of external performance into discrimination at every ordinal threshold and the mapping from score to grade, which shows that the mapping moves in opposite directions on different populations through a shift in the score *given* the true grade;
- an account of what changing the decision rule can and cannot recover, including a negative result for label-free prior adaptation;
- the *matched decision rule*, a comparison rule under which both models' cut-points are refitted by the same procedure, with its record of changed, unchanged and reversed comparisons;
- the number of locally labelled referable cases needed to set a referral threshold, with a distribution-free probability of harm that translates into images at any screening prevalence, and what the operating point means per 10 000 people screened.

2. Materials and methods

2.1. Study design and analysis plans

This is a retrospective modelling study on public, de-identified fundus photographs. An evaluation protocol (unit of independence, held-out sets, model selection on development data only, metrics and uncertainty) was fixed before the first training run. For the analyses reported here, the plan — hypothesis, control, analysis and the result that would falsify it — was committed to the public repository before the analysis was run: the derivation test of the macular-oedema grade (Supplementary Note 2), nine first-stage outcomes O1–O9 (seed replicates, reverse direction, second external corpus, five-fold foundation model) and second-stage outcomes A1–E2 (mechanism, label-free rules, sample size, two further external sets and the models trained on EyePACS), written after an internal pre-submission review of an earlier draft. These are analysis plans committed to version control, not entries in a public registry; their commit identifiers and times are in Supplementary Table S1, and every outcome with its prediction, falsifying result and verdict is in Supplementary Table S2. Two limits apply. The repository history begins on 31 August 2026, so the pre-specification of development runs trained before that date cannot be verified, and we call those models pre-specified only in the sense that the set of ten was fixed in the first-stage plan. And the second-stage prediction A1 was written knowing the overall direction of the shift on APTOS and DDR; what it added was the grade-by-grade form, and DeepDRiD and SUSTech-SYSU, for which no direction was known or predicted, were its first test on new populations. Analyses added after an internal editorial assessment of the manuscript are labelled *post hoc*. Reporting follows TRIPOD+AI[14]

and CLAIM[15] (Supplementary Table S3).

2.2. Data

The *development pool* combined IDRiD[16, 17] (516 images) and Messidor-2[18, 19] (1,744 gradable images with the adjudicated grades of Krause *et al.*[20]), 2,260 images in total (Table 1; Fig. 1). Messidor-2 images were taken from a public mirror of the original release preprocessed to 512×512 pixels; the two higher-resolution models (560 and 640 pixels) used a mirror at the original 2240×1488 pixels that covers 1 057 of the 1 744 images (Supplementary Table S4). The external sets, never used for training, model selection or threshold fitting, were APTOS 2019 (3,662 images, India, single grader), DDR[21] (12,522 gradable images, 147 hospitals in China, seven graders; used from a public 512-pixel mirror), the regular fundus images of DeepDRiD[22] (2,000 images of 500 patients from three Chinese diabetes screening programmes) and SUSTech-SYSU[23] (1,151 graded images; the 68 images of laser scars carry no grade). SUSTech-SYSU is a case–control collection: its grade-0 images are of healthy volunteers without diabetes, photographed at routine health examinations, and its DR images were graded by two ophthalmologists with adjudication by a third[23]. In the reverse direction APTOS was the training corpus and Messidor-2 and IDRiD were external. The EyePACS 2015 training set (35,126 images, single grader) was used to pretrain the development models and, in the second stage, as the development set of two further models. All corpora use the international 5-grade scale[24]; referable DR is grade $\geq$2 and, because macular oedema is not graded in the external sets, it is defined by DR grade alone. A content audit (256-bit difference hash after the training crop) checked every devel-

opment and EyePACS image against every external image (Supplementary Note 7). Patient identifiers are published only for DeepDRiD and EyePACS; Messidor-2 is distributed as pairs of images per examination, but the pairing file is not part of the mirrors used, so resampling elsewhere is by image (near-duplicates merged).

2.3. Models

All models share one design: a backbone, global average pooling and ordinal threshold heads that output four logits for $P(y > k)$[25], trained with masked binary cross-entropy, a grade-balanced sampler, AdamW, cosine decay, weight averaging and dihedral test-time augmentation (details in Supplementary Table S4). The reference model is a DenseNet121[26] pretrained on ImageNet and then EyePACS and fine-tuned at 448 pixels; the selected model replaces the backbone with EfficientNet-B3[27]. The other eight pre-specified development models change one factor each (Table 2): resolution (224, 560 and native 640 pixels), a macula-centred crop, linear-probe-then-fine-tune[28], CORAL and CORN heads[25, 29] and ConvNeXt-tiny[30]; the retinal foundation model RETFound[31] was fine-tuned separately. Development results are out of fold over a frozen 5-fold split stratified by corpus and grade; external predictions average the five fold models. The development models had seen EyePACS during pretraining; what the two EyePACS models (DenseNet121 and EfficientNet-B3, 448 pixels, 15 epochs from ImageNet) changed was the final training set and its labels. They were trained on four folds of a patient-grouped split of EyePACS (28,096 images) and used the fifth (7,030 images) both to select the checkpoint and to fit development rules, so their development estimates are single-fold and slightly optimistic.

2.4. Decision rules and the matched decision rule

For an ordinal model with $p_k = P(y > k)$ the expected grade is $s = \sum_k p_k$. Every model's outputs were decoded under the same rules (Supplementary Note 1). **R1** is the shipped rule: the number of leading thresholds with $p_k > 0.5$. **R2** places four cut-points on s to maximise QWK on development out-of-fold predictions, and **R3** moves R2's grade-1/2 cut to give 90% development referable sensitivity. **R4** and **R4m** fit the cut-points on the external labels, maximising QWK or macro-averaged recall, by two-fold cross-fitting; they are diagnostics of what local labels could recover, never performance. **R5** adapts the class posteriors (derived from the ordinal outputs) to the external population without labels by EM estimation of the target prior[32], and **R5b** by black-box shift estimation[33]; both decode by argmax, so their like-for-like reference is **R0a**, the argmax of the unadapted posteriors.

The *matched decision rule* is the comparison rule used throughout: no difference between two models is attributed to the representation unless it survives refitting both models' cut-points by the same cross-fitted procedure (on development data, fold f's cut-points come from the other four folds; externally, R4). Because a per-grade verdict can depend on how the cut-points are cross-fitted (Supplementary Note 3), every matched claim names its procedure.

2.5. Decomposing external performance

Discrimination was measured at every ordinal threshold (area under the ROC curve of p_k for $y > k$, $k = 0, \ldots, 3$) and overall by Harrell's C of s. The mapping from score to grade was described by the mean expected grade given the true grade, $\mathrm{E}[s \mid y = k]$; under a pure prior (label) shift this

quantity is the same in every population, so a grade-by-grade departure from its development value indicates a conditional shift (outcome A1). To test whether image quality alone can produce the under-grading seen on DDR (A2), the development pool was re-scored out of fold after five fixed degradations of the cached image (JPEG quality 30 at 512 pixels; downsampling to 256 pixels; Gaussian blur, $\sigma = 2$ pixels; gamma 1.8 darkening; contrast reduced to 60%), and per-corpus image statistics were computed (A3).

2.6. Threshold transfer and local recalibration

A referral threshold on $P(y \geq 2)$ was fitted on development predictions for target sensitivities of 80%, 85% (the superiority endpoint of the first autonomous pivotal trial[4]), 87.2% and 95.5%, and applied unchanged externally.

Local recalibration was sized with the operational objective. In the prespecified analysis (C), n external images ($n = 25$–800, 500 draws each) were drawn at random, the referral threshold was set to the largest value giving at least 85% sensitivity on those n, and sensitivity was measured on the remaining images; a draw counts as *harm* when that sensitivity falls below 80%. Because this threshold depends only on the referable images in the sample, and the external sets are enriched (40–45% referable), the analysis was repeated post hoc in terms of the number k of referable cases labelled ($k = 20$–300, 2 000 draws), with the threshold's sensitivity measured on all of the corpus's referable images not drawn. With continuous scores the population sensitivity of the cut at the j-th largest of k referable scores ($j = \lceil 0.85k \rceil$) follows a Beta(j, $k - j + 1$) distribution whatever the model or population, which gives the risk of harm in closed form; for n consecutive

images at referable prevalence π the number of referable cases is binomial(n, π). Positive and negative predictive values and the number of referable patients missed per 10 000 screened were derived at referable prevalences of 5, 10 and 20%.

2.7. Statistics

The primary grading metric is quadratic weighted kappa (QWK)[34]; the operationally primary pair is referable sensitivity and specificity, never quoted alone. Per-grade recall, precision and F1, Cohen's κ, macro-averaged recall (a diagnostic and the objective of R4m, never a deployment objective), area under the curve, Brier score and expected calibration error are reported beside them. Intervals are 95% percentile intervals from 2 000 bootstrap resamples of groups (patients where identified, otherwise images); differences between models are paired bootstraps, and an interval containing zero is reported as indistinguishable, not ranked. Model rankings were compared with Kendall's τ, with a two-sided permutation p-value (10 000 permutations) and, post hoc, a 95% interval from 1 000 image resamples. Because case mix alone can raise an area under the curve, the referral AUC was also computed post hoc with each image weighted to the development grade mix (within referable and within non-referable images), beside the AUC between adjacent grades. A difference between configurations is treated as beyond training noise only if it exceeds $2\sqrt{2}$ standard deviations of matched development QWK across five seeds of the reference model. No correction for multiplicity is applied; per-grade claims are treated as credible only when they replicate across conditions.

2.8. Software and reproducibility

Every number in the article is generated from archived per-image predictions into a ledger from which the manuscript is compiled; no number is typed by hand. An AI assistant (Claude, Anthropic) was used under the authors' direction to write analysis and training code, to run the analyses, to submit the training and inference jobs to Kaggle and to draft text. The authors directed the work and its interpretation, reviewed the code and its outputs, verified the reported results against the archived predictions and take responsibility for them. Code, splits, analysis plans and predictions are public (Data and code availability).

3. Results

3.1. Cohorts and models

The ten pre-specified development models reached a matched development QWK between 0.8267 and 0.8954; the selected EfficientNet-B3, chosen on development data before any external number was examined, reached 0.8954 (Table 2). The four external populations differed in their grade distributions (Table 1): referable DR made up 40.6% of APTOS, 44.9% of DDR, 43.2% of DeepDRiD and 43.1% of SUSTech-SYSU.

3.2. Ranking largely transferred to new populations; the mapping from score to grade did not

Discrimination largely survived the change of population (Fig. 2a). Across the ten development models the area under the ROC curve for referable disease, $P(y > 1)$, was 0.914–0.973 on APTOS, 0.921–0.955 on DDR and

0.943–0.969 on DeepDRiD, against 0.952–0.977 in development; Harrell's C of the expected grade was 0.948–0.959 on APTOS and 0.889–0.923 on DDR, against 0.921 for the reference model in development. Ranking did not survive everywhere: the proliferative threshold, weak already in development (0.814–0.963), fell to 0.783 at worst, and the foundation model's referral ranking degraded on DDR (0.855; below).

On SUSTech-SYSU the referral area under the curve was 0.996–0.999, higher than in development, which better models cannot explain (Supplementary Table S12). The set is a case–control collection whose grade-0 images come from healthy volunteers without diabetes, and only 24 of its images (2.1%) are mild. Weighting its images to the development grade mix left the area under the curve at 0.984–0.997, and no-DR against moderate images separated almost perfectly (0.998–1.000, against 0.939–0.979 in development), so the excess lies in how clear-cut the cases are within grades, not only in the grade mix. APTOS showed a narrower version of the same pattern: its no-DR and mild images separated far better than in development (0.973–0.996 against 0.664–0.815), while its mild and moderate images separated worse (0.534–0.800). Acquisition alone carried grade information in these sets (Supplementary Table S12): the width of an APTOS image separated images with any DR from those without at an area under the curve of 0.862, and in SUSTech-SYSU, whose images all share one size, the estimated JPEG quality of the file separated referable images at 0.670 and sharpness, which lesions also raise, at 0.789. Models resize every image, but cameras that differ in resolution differ in ways a model can see, so part of this excess discrimination may reflect acquisition rather than retinopathy. We therefore do not count

SUSTech-SYSU as evidence that ranking transferred; that evidence rests on APTOS, DDR and DeepDRiD.

Where the scores fell did not transfer, and it moved in opposite directions (Fig. 2b). On APTOS and SUSTech-SYSU the models scored images of the same true grade higher than in development for at least three of the five grades (APTOS: 10 of ten models; SUSTech-SYSU: 10 of ten); on DDR and DeepDRiD they scored them lower (DDR: 10 of ten; DeepDRiD: 10 of ten; pre-specified outcome A1 held). The overall direction on APTOS and DDR was known when A1 was written; on DeepDRiD and SUSTech-SYSU it was not, and the same conditional pattern appeared. The shift is therefore not a change in prevalence alone, under which the score given the true grade would not move. Its shape differed between the two populations. On APTOS it was concentrated in one grade: for the reference model, mild images had a mean expected grade of 1.90 against 0.83 in development, while no-DR images scored lower (0.19 against 0.36). On DDR every grade scored lower, from 0.72 against 0.83 for mild to 2.75 against 3.05 for proliferative DR. Over-grading on APTOS and under-grading on DDR followed: the mean predicted-minus-true grade at the shipped rule was $+0.14$ to $+0.26$ on APTOS and -0.35 to -0.17 on DDR.

A referral threshold fitted on development predictions therefore did not hold its operating point (Fig. 3). Fitted for 85% sensitivity, the reference model referred 99.33% of referable patients on APTOS and 98.59% on SUSTech-SYSU but 63.38% on DDR (pre-specified O8 held); only on DeepDRiD, where the shift was smallest, did it land near its target (85.07%).

Models fine-tuned on 28,096 EyePACS images instead of the development

pool reproduced the direction of the shift but not the sign of the threshold error (Supplementary Table S7). Like the development-pool models, they over-graded APTOS (mean predicted-minus-true grade +0.20 for DenseNet121, +0.14 for EfficientNet-B3) and under-graded DDR (−0.18 and −0.23), with a referral area under the curve of 0.938–0.999 on the six external sets. Yet a threshold fitted on their held-out EyePACS fold for 85% sensitivity over-referred on every set (90.8–100.0%; outcome E2 held), DDR included (90.9% and 90.8%, against 63.38% from the development pool). A transferred threshold carries the reference standard of the set it was fitted on. EyePACS referable images, each graded once, were harder to rank than those of any external set (development area under the curve 0.917 and 0.917, against at least 0.938 externally), so the cut that catches 85% of them is permissive elsewhere; the adjudicated development pool gives a stricter cut, which under-refers where the models under-grade. The same reference standard is the likely reason for the modest development QWK of the EyePACS models (0.7503 and 0.7747). The direction of the shift belonged to the target population; the sign of a transferred threshold's error belonged to the development labels.

3.3. Why the mapping moved

Image quality can produce under-grading, but it does not explain DDR. Re-scored out of fold after degradation, the development pool was under-graded by both re-scored models: JPEG compression at quality 30 shifted the mean predicted-minus-true grade by −0.45 and −0.53 and referable sensitivity at the shipped rule by −36.8 and −45.6 points; Gaussian blur shifted the reference model's mean grade by −0.30 and its sensitivity by −15.4 points (outcome A2 held; the undegraded re-scoring reproduced the archived out-

of-fold QWK, 0.868 against 0.860; Supplementary Table S8). DDR images, however, were not degraded on these measures: as used here they share the 512-pixel size and the estimated JPEG quality of the Messidor-2 development images (95 against 95) and have higher contrast and Laplacian variance (70 against 39), and no acquisition statistic separated DDR's referable images better than 0.597. Resolution, compression and blur therefore do not account for the under-grading on DDR; camera and population characteristics we could not measure, or a stricter grading convention, remain candidates that public data cannot separate. The APTOS pattern is of a different kind: mild images alone moved up by about one grade while no-DR images moved down, which a global change in image quality does not produce and a difference in how APTOS's single grader assigned the mild grade would. Where that is the cause, local recalibration fits the local grading standard (Discussion).

3.4. What the decision rule can and cannot recover

Where ranking survived, much of the external loss was a property of the rule. On APTOS every development model lost the mild grade at the shipped rule (recall 1.6–14.3%, against 26.8–62.0% in development; Fig. 4a), and so did all five seeds of the reference model (5.4–6.2%; O3 held). Cut-points refitted on half of the external labels and applied to the other half (R4m) raised mild recall to 61.6–77.8% with mild F1 of 0.501–0.544 (precision 0.406 for the selected model), above the development mild F1 of the same models under the same macro-recall objective with cut-points cross-fitted across folds (0.262–0.420; 0.395 for the selected model). The gain was a trade, not a free repair: 342 moderate images were relabelled mild and referable sensitivity fell from 99.3% to 75.3%, because mild DR is not referable. In the reverse

direction the model trained on APTOS under-graded Messidor-2 (O6 held) and referred 43.5% of referable eyes at its shipped rule while ranking them with an area under the curve of 0.890; refitted cut-points raised sensitivity to 77.9%.

Where ranking or the grade itself did not survive, the rule could not help. On DDR the mild grade was not collapsed (recall 20.5–45.2%) and no rule raised its F1 above 0.235. The collapse of mild recall was specific to APTOS, but low mild F1 was not: even with cut-points refitted on external labels it stayed at or below 0.330 on DeepDRiD and 0.305 on SUSTech-SYSU, as in the released ORDER-DR predictions (below) and in a leave-one-domain-out benchmark of ordinal models in which mild F1 never exceeded about 0.16 on any held-out domain[13]. In the reverse direction mild recall at the shipped rule was 34.4%, so the pre-specified predictions that a fine-tuned model would reproduce the published mild-grade collapse (O4) and that refitted cut-points would triple it (O5) were both falsified. RETFound ranked APTOS images as well as the convolutional models (area under the curve 0.947) and most of its QWK deficit there was the rule (0.703 at its shipped rule, 0.834 refitted), but on DDR its ranking itself degraded (0.855; QWK 0.646 even refitted).

Rules that need no external labels did not close the gap. Cut-points fitted on development data (R2, R3) left APTOS mild recall at 5.7–22.7% and lowered external QWK for 9 of 10 models. Label-free prior adaptation (R5) closed at least a quarter of the DDR referral-sensitivity gap between R1 and R4 in only 2 of 10 models (outcome B was falsified). Because R5 decodes by argmax, its fair reference is the argmax of the unadapted posteriors (R0a), which already differs from R1; against R0a, prior adaptation changed referable

sensitivity by a median of +0.2 points on APTOS, +1.5 on DDR, +2.4 on DeepDRiD and +0.1 on SUSTech-SYSU, and predicted any mild image in only 7 of 40 model–population pairs (Supplementary Table S11). That is what a shift in the score given the grade, rather than in prevalence, should produce.

3.5. Model comparisons depend on the decision rule

The pre-specified prediction that the choice among deployable rules (R1–R3) would move mild recall and referable sensitivity more than the choice among models (O7) was falsified on both APTOS and DDR: models trained on the same data differed materially at their shipped rules (median within-model range 5.7 against a between-model range of 12.7 points of mild recall on APTOS, 7.1 against 11.4 points of referable sensitivity on DDR). Which model looked best, however, depended on the rule and on the metric. The rankings of models by external QWK at the shipped rule and by referable sensitivity at a fixed 90% specificity were at most weakly related (Kendall τ −0.16 [−0.33, +0.20] on APTOS, +0.22 [+0.13, +0.42] on DDR, +0.22 [+0.18, +0.72] on DeepDRiD and +0.53 [+0.27, +0.59] on SUSTech-SYSU; permutation p 0.07–0.61), so with ten models neither ranking could be predicted from the other.

Under the matched decision rule several conclusions of the usual kind changed (Table 3). On APTOS, the selected EfficientNet-B3 model led the reference DenseNet121 by +0.0059 QWK [+0.0007, +0.0110] at the shipped rule and by +0.0032 [−0.0020, +0.0083] with both models refitted, and per-grade differences changed sign between operating points. On development data, linear-probe-then-fine-tune appeared to cost −5.54 accuracy points

[−7.48, −3.54] at shipped cut-points and gained +1.95 [+0.18, +3.72] at matched ones. The rule is not a one-way filter: the development advantage of EfficientNet-B3 over its DenseNet121 control survived it (+0.0202 shipped, +0.0207 [+0.0105, +0.0320] matched).

The same held for an independently trained, published model family. Applying our rules to the released image-level probabilities of ORDER-DR[11] reproduced its reported Messidor-2 result exactly (QWK 0.6423 ± 0.0364; Supplementary Table S6); with external cut-points all four released models converged to QWK 0.672–0.683, so the reported advantage of the fused model over its best branch was a decision-rule effect, while mild F1 stayed at or below 0.285.

Training noise set the scale for all of these effects (O1). Matched development QWK varied with a standard deviation of 0.0043 across five seeds of one configuration (0.8646–0.8764). Between models that differed only in their seed, 15 of 50 per-grade recall comparisons on APTOS had a bootstrap interval excluding zero at the shipped rule and 29 with refitted cut-points (O2 held). An image-level bootstrap captures the sampling of test images, not the variance of training, so per-grade "effects" of the size usually attributed to architecture arise between identical architectures.

3.6. Agreement statistics cannot see the referral operating point

Holding the other cut-points at their QWK-optimal values and sweeping the referral cut (Fig. 5), every operating point of the selected model within 0.01 of its maximum QWK on DDR spanned referable sensitivity from 71.3% to 86.3%: between 773 and 1,615 of 5,626 referable patients missed, with QWK unable to tell the operating points apart. In a screening programme

with 10% referable prevalence, the selected model's shipped rule on DDR would miss 297 referable patients per 10 000 screened (negative predictive value 96.8%), and cut-points refitted on local labels 188, at the cost of 521 rather than 117 false referrals (Supplementary Table S9).

3.7. How many local labels a referral threshold needs

With the pre-specified objective — a referral threshold set on n locally labelled images for 85% sensitivity — the probability that sensitivity on the remaining images fell below 80% with 200 images was 9.4–15.8% for the selected and reference models on every external set (outcome C held), while mean sensitivity stayed close to target at every size. Those sets are enriched, however (40.6–44.9% referable): 200 of their images contain about 86 referable cases, and the threshold depends on nothing else. Counted in referable cases, the risk was nearly identical on every population and for both models and agreed with the distribution-free closed form within 3.5 points (Fig. 6a; Supplementary Table S10): 41.1% with 20 referable cases, 21.3% with 60, 10.2% with 120 and 4.3% with 200. At a screening prevalence of 10%, 200 consecutive images therefore carry a 31.6% risk, and about 1 000 (10.5%) to 2 000 images (3.6%) are needed (Fig. 6b). Fewer cases suffice if the threshold is set more conservatively: with 60 referable cases, a cut at 90.0% sample sensitivity keeps the risk at or below 5%, at a cost in specificity that depends on the population (for the selected model on DDR, 88.1% rather than 93.8% at the thresholds giving 90% and 85% sensitivity).

3.8. Interventions on the representation

Of 9 pre-specified interventions on the model, 8 failed their criterion under the matched decision rule (Supplementary Fig. S1 and Table S5). Only the EfficientNet-B3 backbone improved development QWK; the retinal foundation model, fine-tuned at its native 224 pixels, fell below a DenseNet121 at the same resolution by -0.0414 [-0.0583, -0.0238] over 5 folds (O9 held). No intervention moved the macular-oedema head, whose middle grade is supervised by 51 images (Supplementary Note 2).

4. Discussion

In the DR-grading settings studied here, an external failure separated into two components that call for different remedies. The ranking of images by severity largely transferred to new populations, also for models whose final training set was EyePACS rather than the development pool; the mapping from score to grade did not, and its error had opposite signs on different populations because the score given the true grade moved. Where ranking survived, part of the loss was a property of the decision rule and could be recovered without retraining — the mild grade on APTOS, referral sensitivity in the reverse direction — but only as a trade between grades, and only with local labels. Where ranking degraded, as for the foundation model on DDR, no rule helped. Two of our own predictions bound the argument: deployable rules did not outweigh the choice of model, and the collapse of mild recall that the literature reads as representational blindness was specific to one corpus, although mild F1 stayed low on every population. A reader who sees an external failure should therefore ask, in order, whether the model still ranks

the new population, where its cut-points were set and on which population, and only then what it has learned.

These results refine rather than contradict recent work. ORDER-DR separated threshold-dependent grade assignment from threshold-independent risk ranking and noted threshold shift on external data[11]; reanalysing its released predictions shows that its reported advantage over its own branches does not survive the matched decision rule. Tabibu and Abhyankar found, in a preprint, that the probability calibration of a binary referral model degrades across cameras mostly through prevalence, and that 100–200 labelled images restore it[12]. The mechanism we found differs: the score given the true grade moved, so prior adaptation without labels failed even against its own decoding, and temperature-style rescaling cannot be expected to help either, as reported for frozen encoders[8]. For a referral threshold, the number of labels needed is set by the referable cases among them, and the risk follows a distribution-free form, so a few hundred images suffice only where referable disease is common. A public, not yet peer-reviewed model repository reports that per-threshold calibration degrades unevenly under domain shift and that the mild grade fails on every held-out domain[13], consistent with the grade-specific movement and the low mild F1 we observed.

The central objection to this kind of analysis is that refitting cut-points on local labels may fit local label noise rather than fix the model. The populations answer it differently. On DDR every grade moved down by a similar amount, yet the images were not degraded on any measure we could compute, although degradation can produce exactly this pattern; the change lies in what the images show or in how they were graded, and recalibration

corrects the mapping without revealing which. On APTOS one grade moved by about a whole step while its neighbours did not, which a change in image quality does not produce and a difference in how a single grader assigned the mild grade would; recalibration there aligns the model with the local grading standard. The models trained on EyePACS add a second line of evidence: they under-graded DDR as the development-pool models did, yet their EyePACS-fitted threshold over-referred there, because a transferred threshold carries the reference standard of the set it was fitted on. Where a threshold lands is thus partly a property of the labels on both sides, which is consistent with, though not proof of, a difference in grading convention; re-grading a random sample of APTOS mild and DDR images by one ophthalmologist would test it directly. Whether recalibration is a fix depends on whether the local standard is the one the screening programme acts on — which, for a programme deploying a model, it usually is. Either way, the rule must be fitted where it will be used, on a sample large enough for its risk to be known.

The practical recommendations cost no additional training. External evaluations should report discrimination at each ordinal threshold beside grade-level agreement, the full referral operating curve, and the transfer gap of any development-fitted threshold, since the same threshold landed $+18.1$ and -29.0 points from an 80% target on two populations. Model comparisons should be made under a matched decision rule that names its fitting procedure, with differences judged against the variation between seeds of one configuration. Deployment should include setting the referral threshold on a local sample, sized by its referable cases rather than its images: about 120 referable cases keep the probability of missing an 80% sensitivity floor near

one in ten and about 200 near one in twenty, which at a referable prevalence of 10% means 1 000 to 2 000 consecutive images; where fewer cases are available, setting the cut for a higher sample sensitivity buys the same safety with specificity. This is in line with advice on updating prediction models for new settings[9, 10, 35], and because the risk does not depend on the model or the population it can be planned before any local data are collected.

The study has limitations. All data are public and retrospective, the populations are defined by dataset rather than by a prospective screening programme, and APTOS and EyePACS labels come from single graders. SUSTech-SYSU is a case–control collection with few mild images, and APTOS's no-DR and mild images separate unusually easily, partly through acquisition, so discrimination there overstates what a screening population would show; acquisition statistics also carried grade information within the development pool, whose two corpora differ in both storage and grade mix (estimated JPEG quality, area under the curve 0.790), a shortcut the models may have used. Patient identifiers are unavailable for most corpora, so resampling is by image and fellow-eye pairs may be correlated. The external-label rules (R4, R4m) are diagnostics of what local labels could recover, not external validation; the deployable recommendation rests on the separate sample-size analysis. Referable disease was defined by DR grade alone because macular oedema is not graded in the external sets, whereas real screening refers for both; the macular-oedema head could not be validated externally and is reported only in the Supplement. The reverse direction, the seeds and the large-data models each use one or two architectures, and the EyePACS models selected their checkpoint on the same fold that set their development thresh-

old. Finally, the analysis plans were committed to version control rather than to a public registry, the pre-specification of development runs trained before the repository history begins cannot be verified, the second-stage prediction on the direction of the shift was written knowing it for APTOS and DDR, and the analyses added after an internal editorial assessment, including the sample-size analysis in referable cases, are post hoc. The decision-rule effects should be confirmed prospectively, with cut-points fitted on an initial local sample and then frozen.

5. Conclusions

External failures and apparent differences between DR grading models are routinely attributed to learned representations. In the settings studied here a substantial part of them lay in the decision rule that converts scores into grades and could be diagnosed and repaired without retraining, while the remainder reflected real losses of ranking that no rule recovers. Reporting per-threshold discrimination, full operating curves, the transfer gap of thresholds and matched-decision-rule comparisons, and setting referral thresholds on a local sample with enough referable cases before deployment, would separate the two.

CRediT authorship contribution statement

Ahmadreza Nourozi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing, Project administration. **Alireza Chegeni:**

Conceptualization, Methodology, Investigation, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Ethics statement

The study used only publicly available, de-identified fundus photographs released for research (IDRiD, Messidor-2, APTOS 2019, EyePACS 2015, DDR, DeepDRiD and SUSTech-SYSU). No new data were collected from human participants, and ethical approval was not required.

Data and code availability

All images are from public datasets. Code, the frozen splits, per-image out-of-fold and external predictions for every model, the analysis plans with their commit history, and every generated table and figure are available at `https://github.com/Ahmadrezanourozii/Automatic-detection-of-diabetic-retinopathy-and-grading-of-diabetic-macular-edema-using-CNN`.

Acknowledgements

We thank the creators of IDRiD, Messidor-2 (and Krause *et al.* for its adjudicated grades), APTOS 2019, EyePACS, DDR, DeepDRiD and SUSTech-SYSU for releasing their data, and Sheng *et al.* for releasing the image-level predictions that made the reanalysis of ORDER-DR possible. Computing used Kaggle's free GPU allocation.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to draft and edit text and, as described in Section 2 (Software and reproducibility), to write and run analysis code and to submit computing jobs. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

[1] Z. L. Teo, Y.-C. Tham, M. Yu, M. L. Chee, T. H. Rim, N. Cheung, et al., Global prevalence of diabetic retinopathy and projection of burden through 2045: systematic review and meta-analysis, Ophthalmology 128 (11) (2021) 1580–1591. doi:10.1016/j.ophtha.2021.04.027.

[2] V. Gulshan, L. Peng, M. Coram, M. C. Stumpe, D. Wu, A. Narayanaswamy, et al., Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs, JAMA 316 (22) (2016) 2402–2410. doi:10.1001/jama.2016.17216.

[3] D. S. W. Ting, C. Y.-L. Cheung, G. Lim, G. S. W. Tan, N. D. Quang, A. Gan, et al., Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes, JAMA 318 (22) (2017) 2211–2223. doi:10.1001/jama.2017.18152.

[4] M. D. Abràmoff, P. T. Lavin, M. Birch, N. Shah, J. C. Folk, Pivotal trial of an autonomous AI-based diagnostic system for detection of diabetic retinopathy in primary care offices, npj Digital Medicine 1 (2018) 39. doi:10.1038/s41746-018-0040-6.

[5] E. Ipp, D. Liljenquist, B. Bode, V. N. Shah, S. Silverstein, C. D. Regillo, et al., Pivotal evaluation of an artificial intelligence system for autonomous detection of referrable and vision-threatening diabetic retinopathy, JAMA Network Open 4 (11) (2021) e2134254. doi:10.1001/jamanetworkopen.2021.34254.

[6] A. Y. Lee, R. T. Yanagihara, C. S. Lee, M. Blazes, H. C. Jung, Y. E. Chee, et al., Multicenter, head-to-head, real-world validation study of seven automated artificial intelligence diabetic retinopathy screening systems, Diabetes Care 44 (5) (2021) 1168–1175. doi:10.2337/dc20-1877.

[7] M. Duggal, A. Chauhan, V. Gupta, A. Kankaria, D. Budhija, P. Verma, V. Miglani, P. Syal, G. Kaur, L. Kumar, N. Mutyala, R. Bezbaruah, N. Sood, A. Kernohan, G. Menon, L. Vale, Real-world evaluation of AI-driven diabetic retinopathy screening in public health settings: validation and implementation, JMIR Medical Informatics 13 (2025) e67529. doi:10.2196/67529.

[8] M. Poyrazer, H. Yağcı, R. Erten, How well do frozen foundation models transfer? A calibration-focused benchmark for diabetic retinopathy grading, Frontiers in Medicine 13 (2026) 1815982. doi:10.3389/fmed.2026.1815982.

[9] K. J. M. Janssen, K. G. M. Moons, C. J. Kalkman, D. E. Grobbee, Y. Vergouwe, Updating methods improved the performance of a clinical prediction model in new patients, Journal of Clinical Epidemiology 61 (1) (2008) 76–86. doi:10.1016/j.jclinepi.2007.04.018.

[10] Y. Vergouwe, D. Nieboer, R. Oostenbrink, T. P. A. Debray, G. D. Murray, M. W. Kattan, et al., A closed testing procedure to select an appropriate method for updating prediction models, Statistics in Medicine 36 (28) (2017) 4529–4539. doi:10.1002/sim.7179.

[11] Q. Sheng, Q. Dong, S. Wu, W. Liang, ORDER-DR: external validation of severity grading and referable-risk stratification from fundus images, Frontiers in Endocrinology 17 (2026) 1923216. doi:10.3389/fendo.2026.1923216.

[12] S. Tabibu, M. Abhyankar, Discrimination transfers but calibration does not: prevalence-dominated miscalibration of diabetic retinopathy screening across fundus cameras, Research Square (preprint, version 1, posted 12 August 2026; not peer reviewed; under review at Scientific Reports, revision requested 12 October 2026)doi:10.21203/rs.3.rs-10323501/v1.

[13] A. F. Rashid, GradEye: CORN ordinal diabetic retinopathy grading checkpoints with leave-one-domain-out evaluation (model repository),

Hugging Face, https://huggingface.co/gradeye; code at https://github.com/ahmed-farhanur-rashid/gradeye, not peer reviewed; accompanying paper not yet available. Accessed 21 October 2026 (2026).

[14] G. S. Collins, K. G. M. Moons, P. Dhiman, R. D. Riley, A. L. Beam, B. Van Calster, et al., TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods, BMJ 385 (2024) e078378. doi:10.1136/bmj-2023-078378.

[15] A. S. Tejani, M. E. Klontzas, A. A. Gatti, J. T. Mongan, L. Moy, S. H. Park, et al., Checklist for artificial intelligence in medical imaging (CLAIM): 2024 update, Radiology: Artificial Intelligence 6 (4) (2024) e240300. doi:10.1148/ryai.240300.

[16] P. Porwal, S. Pachade, R. Kamble, M. Kokare, G. Deshmukh, V. Sahasrabuddhe, F. Meriaudeau, Indian diabetic retinopathy image dataset (IDRiD): a database for diabetic retinopathy screening research, Data 3 (3) (2018) 25. doi:10.3390/data3030025.

[17] P. Porwal, S. Pachade, M. Kokare, G. Deshmukh, J. Son, W. Bae, et al., IDRiD: diabetic retinopathy – segmentation and grading challenge, Medical Image Analysis 59 (2020) 101561. doi:10.1016/j.media.2019.101561.

[18] E. Decencière, X. Zhang, G. Cazuguel, et al., Feedback on a publicly distributed image database: the Messidor database, Image Analysis & Stereology 33 (3) (2014) 231–234. doi:10.5566/ias.1155.

[19] M. D. Abràmoff, J. C. Folk, D. P. Han, J. D. Walker, D. F. Williams, S. R. Russell, et al., Automated analysis of retinal images for detection of referable diabetic retinopathy, JAMA Ophthalmology 131 (3) (2013) 351–357. doi:10.1001/jamaophthalmol.2013.1743.

[20] J. Krause, V. Gulshan, E. Rahimy, P. Karth, K. Widner, G. S. Corrado, et al., Grader variability and the importance of reference standards for evaluating machine learning models for diabetic retinopathy, Ophthalmology 125 (8) (2018) 1264–1272. doi:10.1016/j.ophtha.2018.01.034.

[21] T. Li, Y. Gao, K. Wang, S. Guo, H. Liu, H. Kang, Diagnostic assessment of deep learning algorithms for diabetic retinopathy screening, Information Sciences 501 (2019) 511–522. doi:10.1016/j.ins.2019.06.011.

[22] R. Liu, X. Wang, Q. Wu, L. Dai, X. Fang, T. Yan, et al., DeepDRiD: diabetic retinopathy—grading and image quality estimation challenge, Patterns 3 (6) (2022) 100512. doi:10.1016/j.patter.2022.100512.

[23] L. Lin, M. Li, Y. Huang, P. Cheng, H. Xia, K. Wang, J. Yuan, X. Tang, The SUSTech-SYSU dataset for automated exudate detection and diabetic retinopathy grading, Scientific Data 7 (2020) 409. doi:10.1038/s41597-020-00755-0.

[24] C. P. Wilkinson, F. L. Ferris, R. E. Klein, et al., Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales, Ophthalmology 110 (9) (2003) 1677–1682. doi:10.1016/S0161-6420(03)00475-5.

[25] W. Cao, V. Mirjalili, S. Raschka, Rank consistent ordinal regression for neural networks with application to age estimation, Pattern Recognition Letters 140 (2020) 325–331. doi:10.1016/j.patrec.2020.11.008.

[26] G. Huang, Z. Liu, L. van der Maaten, K. Q. Weinberger, Densely connected convolutional networks, Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (2017) 4700–4708doi:10.1109/CVPR.2017.243.

[27] M. Tan, Q. V. Le, EfficientNet: rethinking model scaling for convolutional neural networks, Proc. International Conference on Machine Learning (ICML) (2019) 6105–6114.

[28] A. Kumar, A. Raghunathan, R. Jones, T. Ma, P. Liang, Fine-tuning can distort pretrained features and underperform out-of-distribution, Proc. International Conference on Learning Representations (ICLR).

[29] X. Shi, W. Cao, S. Raschka, Deep neural networks for rank-consistent ordinal regression based on conditional probabilities, Pattern Analysis and Applications 26 (2023) 941–955. doi:10.1007/s10044-023-01181-9.

[30] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, S. Xie, A ConvNet for the 2020s, Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2022) 11976–11986doi:10.1109/CVPR52688.2022.01167.

[31] Y. Zhou, M. A. Chia, S. K. Wagner, M. S. Ayhan, D. J. Williamson, R. R. Struyven, T. Liu, M. Xu, M. G. Lozano, P. Woodward-Court,

Y. Kihara, et al., A foundation model for generalizable disease detection from retinal images, Nature 622 (7981) (2023) 156–163. doi:10.1038/s41586-023-06555-x.

[32] M. Saerens, P. Latinne, C. Decaestecker, Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure, Neural Computation 14 (1) (2002) 21–41. doi:10.1162/089976602753284446.

[33] Z. C. Lipton, Y.-X. Wang, A. J. Smola, Detecting and correcting for label shift with black box predictors, Proc. 35th International Conference on Machine Learning, PMLR 80 (2018) 3122–3130.

[34] J. Cohen, Weighted kappa: nominal scale agreement with provision for scaled disagreement or partial credit, Psychological Bulletin 70 (4) (1968) 213–220. doi:10.1037/h0026256.

[35] R. D. Riley, T. P. A. Debray, G. S. Collins, L. Archer, J. Ensor, M. van Smeden, et al., Minimum sample size for external validation of a clinical prediction model with a binary outcome, Statistics in Medicine 40 (19) (2021) 4230–4251. doi:10.1002/sim.9025.

Table 1: **Cohorts.** DR is graded on the international 5-grade scale in every corpus. No external image was used for training, model selection or threshold fitting. Messidor-2 and DDR were used from public 512-pixel mirrors; SUSTech-SYSU's grade-0 images are of healthy volunteers without diabetes, and its 68 laser-scar images carry no grade.

Role	Corpus	Origin	Images	Grading
Development	IDRiD	India	516	3 graders (published grading)
	Messidor-2	France	1,744	adjudicated by 3 specialists
Pretraining; EyePACS models	EyePACS 2015	USA	35,126	single grader per image
External	APTOS 2019	India	3,662	single grader
	DDR	China, 147 hospitals	12,522	7 graders
	DeepDRiD (regular)	China, 3 diabetes screening programmes	2,000 (500 patients)	doctor annotations (challenge release)
	SUSTech-SYSU	China, Guangdong; case–control	1,151 graded of 1,219	2 ophthalmologists, adjudicated
Reverse direction	APTOS → Messidor-2, IDRiD		3,662 → 2,260	as above

Table 2: **Models.** Each development model changes one factor relative to the reference. Development QWK: out-of-fold DR QWK under the matched decision rule (fold-wise cross-fitted cut-points); for the two EyePACS models, on their single held-out EyePACS fold (7,030 single-graded images, cut-points two-fold cross-fitted within it), which also selected their checkpoint. Repository codes are given for traceability.

Model	Code	Pixels	Factor changed	Dev. QWK
DenseNet121-448 (reference)	E08	448	reference	0.8646
DenseNet121-224	E09	224	resolution	0.8389
DenseNet121-560	E10	560	resolution	0.8749
DenseNet121-640 native	E17NAT	640	native-resolution source images	0.8662
EfficientNet-B3 (selected)	E11FULL	448	backbone (selected model)	0.8954
DenseNet121 + macula crop	E14MAC	448	macula-centred crop for the DME head	0.8595
DenseNet121 LP-FT	E15LPFT	448	linear probe, then fine-tune	0.8655
DenseNet121 CORAL head	E20CORAL	448	CORAL ordinal head	0.8569
DenseNet121 CORN head	E22CORN	448	CORN ordinal head	0.8564
ConvNeXt-T (2 folds)	E21CNXA	448	backbone (2 of 5 folds)	0.8267
RETFound ViT-L/16	I24FULL	224	foundation encoder (not in the O7 set)	0.7976
DenseNet121, EyePACS-trained	EYEDN	448	trained on EyePACS	0.7503
EfficientNet-B3, EyePACS-trained	EYEEB3	448	trained on EyePACS	0.7747

Table 3: **The matched decision rule changes model comparisons.** Differences (first minus second) with 95% paired-bootstrap intervals at the shipped rule and with both models' cut-points refitted by the same procedure (external rows: R4, two-fold cross-fitted on APTOS labels; development rows: fold-wise cross-fitted). *Interval excludes zero. A per-grade verdict that depends on the cross-fitting procedure is reported in Supplementary Note 3.

Comparison	Shipped rule	Matched decision rule	Verdict
EfficientNet-B3 vs DenseNet121, APTOS QWK	+0.0059 [+0.0007, +0.0110]	+0.0032 [−0.0020, +0.0083]	advantage dissolves
recall, moderate (points)	+1.4	+4.0*	significance changes
recall, severe (points)	−1.0	−10.3*	significance changes
recall, proliferative (points)	+8.1*	+6.8*	persists
LP-FT vs fine-tuning, development accuracy (points)	−5.54 [−7.48, −3.54]	+1.95 [+0.18, +3.72]	reverses
EfficientNet-B3 vs DenseNet121, development QWK	+0.0202 [+0.0071, +0.0329]	+0.0207 [+0.0105, +0.0320]	survives

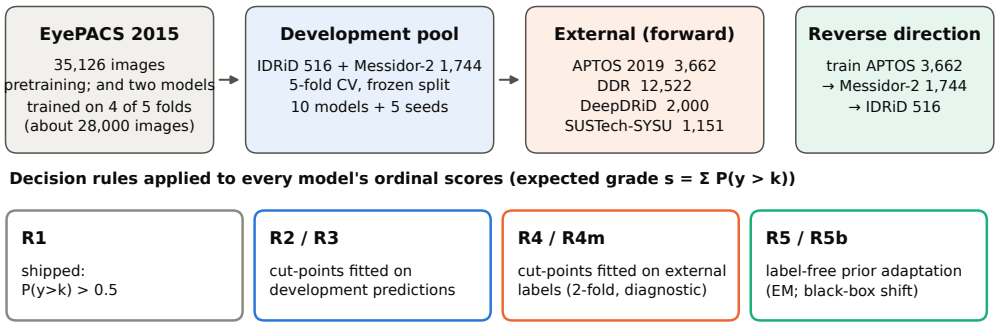


Figure 1: **Study design.** Development models were trained on a pooled IDRiD and Messidor-2 set under a frozen 5-fold split (after pretraining on EyePACS) and evaluated without further tuning on four external populations; a model trained on APTOS was evaluated in the reverse direction, and two models were fine-tuned on 28,096 EyePACS images. Every model's ordinal outputs were decoded under the same decision rules: R1, the shipped default; R2 and R3, fitted on development predictions; R4 and R4m, fitted on external labels by two-fold cross-fitting (diagnostic only); and R5 and R5b, label-free adaptation to the external population, with R0a, the unadapted argmax, as their reference.

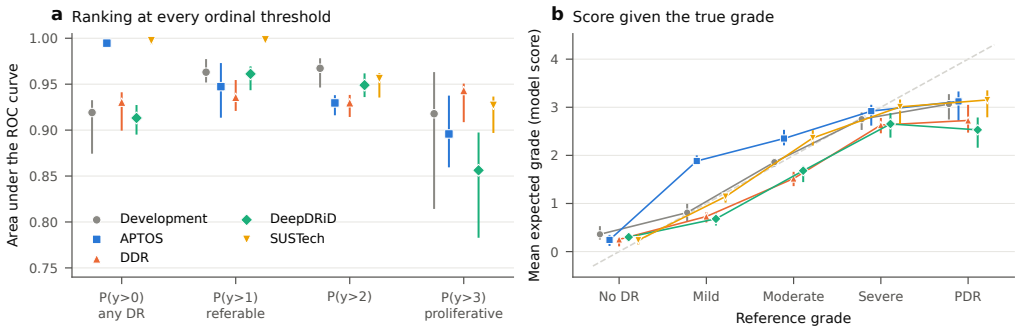


Figure 2: **Ranking transferred; the score given the true grade did not. a**, Area under the ROC curve of each ordinal output $P(y > k)$ in development (out of fold) and in each external population: median (marker) and range (line) across the ten development models. On SUSTech-SYSU, a case–control collection, and for no DR against mild on APTOS, discrimination exceeds development values because of case mix (Supplementary Table S12). **b**, Mean expected grade $\mathrm{E}[s \mid y]$ for images of each reference grade; under a pure change in prevalence these curves would coincide. APTOS images of the mild grade score about one grade higher than in development, whereas DDR images of every grade score lower. Colours are paired with marker shapes.

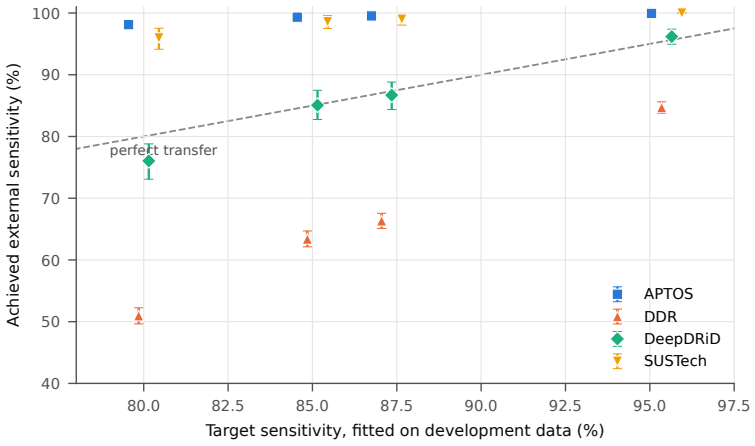


Figure 3: **Referral thresholds fitted on development data do not hold their operating point.** A threshold on $P(\text{grade} \geq 2)$ was fitted on the reference model's development out-of-fold predictions for each target sensitivity (x-axis) and applied unchanged to each external population; points show the achieved sensitivity with 95% bootstrap intervals, and the diagonal is perfect transfer. Targets: 80%; 85% (the superiority endpoint of the first autonomous pivotal trial); 87.2% (that trial's achieved sensitivity); 95.5%. The same thresholds over-refer where the models over-grade (APTOS, SUSTech-SYSU), under-refer where they under-grade most (DDR) and transfer where the shift is small (DeepDRiD).

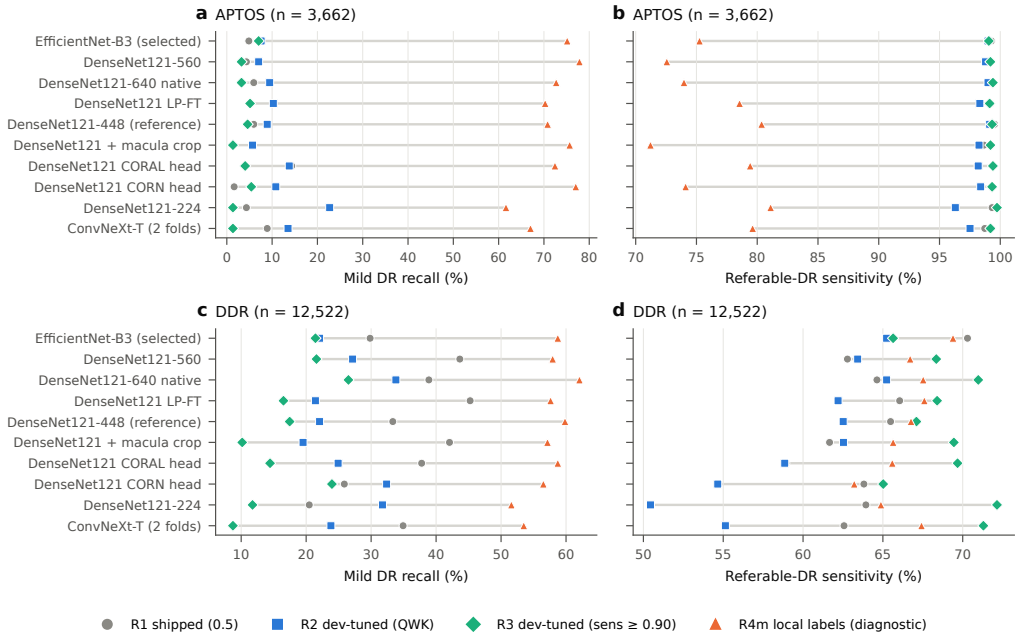


Figure 4: **External mild-grade recall and referral sensitivity of every development model under every decision rule.** Rows are models (Table 2) ordered by development QWK; markers show the same model under four rules. **a**,**b**, APTOS 2019; **c**,**d**, DDR. On APTOS, mild recall at the shipped rule (R1) is low in every model and is restored by cut-points refitted on external labels (R4m, diagnostic) at the cost of referral sensitivity, because mild DR is not referable; rules fitted on development data (R2, R3) do not restore it. On DDR the mild grade is not collapsed and no rule recovers much. Models differ at a fixed rule by more than the deployable rules differ within a model (pre-specified outcome O7 falsified).

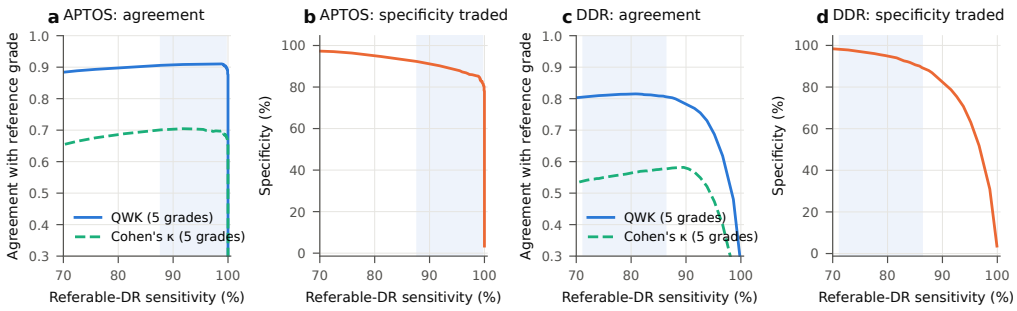


Figure 5: **Agreement statistics cannot see the referral operating point. a**,**c**, QWK and Cohen's κ of the selected model as the referral cut-point is swept with the other cut-points at their QWK-optimal values, against the referable sensitivity it produces, on APTOS (**a**) and DDR (**c**). The shaded band contains every operating point whose QWK is within 0.01 of the maximum. **b**,**d**, the specificity that the same sweep trades for sensitivity.

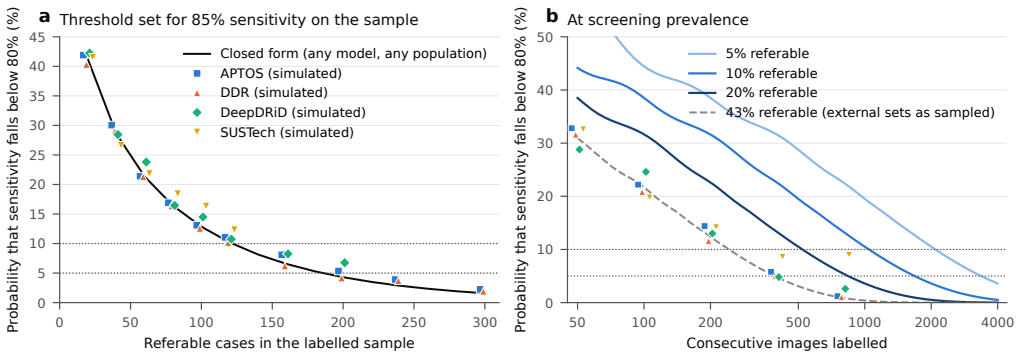


Figure 6: **How many local labels a referral threshold needs.** The threshold is set to the largest value giving at least 85% sensitivity on the referable cases of a labelled sample; harm is a population sensitivity below 80%. **a**, risk of harm against the number of referable cases in the sample: closed form (line), which holds for any model and population with continuous scores, and simulation for the selected model on each external set (markers; 2 000 draws, sensitivity measured on the referable images not drawn; post hoc). **b**, risk against the number of consecutive images labelled at referable prevalences of 5, 10 and 20% (closed form) and at the 40.6–44.9% of the external sets as sampled (dashed), with the pre-specified simulation for the selected model (markers; 500 draws, sensitivity measured on the remaining images, which adds evaluation noise at large n on the smaller sets). Dotted lines, 10% and 5% risk.